

1. Determine todos los valores $z \in \mathbb{C}$ tales que $z^3 = i - 1$, y represéntelos gráficamente. (4 pts)

$$z^3 = i - 1$$

$$z = i - 1$$

$$z = \sqrt[3]{i-1}$$

$$a = -1$$

$$b = 1$$

$$z = \sqrt[3]{2} \cdot \text{cis} \left(\frac{\frac{3\pi}{2} + 2\pi k}{3} \right)$$

$$r = \sqrt{(-1)^2 + 1^2} = \sqrt{2}$$

$$\theta = \arctan \left(\frac{1}{-1} \right) + \pi = \frac{3\pi}{2}$$

$$z = \sqrt{2} \cdot \text{cis} \left(\frac{3\pi}{2} \right)$$

$k=0 \rightarrow$	$\sqrt{2} \cdot \text{cis} \left(\frac{\pi}{2} \right)$
$k=1 \rightarrow$	$\sqrt{2} \cdot \text{cis} \left(\frac{7\pi}{2} \right)$
$k=2 \rightarrow$	$\sqrt{2} \cdot \text{cis} \left(\frac{11\pi}{2} \right)$

2. Sea $p(x) = x^3 - ax^2 + x - a$. Si se sabe que i es raíz doble de $p(x)$ (es decir, i es raíz de multiplicidad dos), halle el valor de a . (3 pts)

$$x^3 - ax^2 + x - a$$

$$\begin{array}{r|l} 1 & -a & 1 & -a \\ & i & -ai-1 & a \\ \hline 1 & -a+i & -ai & 0 \end{array} \quad \begin{array}{l} i \\ -ai-1 \end{array}$$

$$\begin{array}{r|l} x^2 & +(-a+i)x & +(-ai) \\ 1 & -a+i & -ai \\ & -i & ai \\ \hline 1 & -a & 0 \end{array} \quad \begin{array}{l} -i \\ -i \end{array}$$

$$(x - (-i))(x + i)(x - a)$$

$$(x + i)(x + i)(x - a)$$

$$x = -i \quad x = -i \quad x = a$$

$$a = i$$

3. Sean $z = 2 + abi$ y $w = 2a - b + 3i$. ¿Cuál debe ser la relación entre los números reales a y b para que $\text{Re}(zw) = 0$? (3 pts)

$$zw = (2 + abi)(2a - b + 3i)$$

$$2a - 2b + 6i + 2a^2bi - ab^2i - 3ab = 0$$

$$(2a - 2b - 3ab) + (2a^2b + bi - ab^2)i = 0$$

$$7a - 2b - 3ab = 0$$

$$7a - 3ab = 2b$$

$$a(7a - 3b) = 2b$$

$$\boxed{a = \frac{2b}{7a - 3b}}$$

4. Si $w \in \mathbb{C}$, obtenga la forma rectangular de w , donde $w + e^i = 2 \operatorname{Ln}(1 - i)$.

$$w = 2 \operatorname{Ln}(1 - i) - e^i$$

$$z = 1 - i$$

$$2 \operatorname{Ln}(\sqrt{2}) + 2 \operatorname{Ln}(e^{i \frac{-\pi}{4}}) - e^i$$

$$a = 1$$

$$b = -1$$

$$2 \operatorname{Ln}(\sqrt{2}) + 2 \cdot i \frac{-\pi}{4} - e^i$$

$$r = \sqrt{1^2 + (-1)^2} = \sqrt{2}$$

$$2 \operatorname{Ln}(\sqrt{2}) - \frac{i\pi}{2} - e^i$$

$$\theta = \arctan\left(\frac{-1}{1}\right) = \frac{-\pi}{4}$$

$$2 \operatorname{Ln}(\sqrt{2}) - e^i - \frac{i\pi}{2}$$

$$z = \sqrt{2} \cdot e^{i \frac{-\pi}{4}}$$

$$2 \operatorname{Ln}(2)^{\frac{1}{2}} - e^i - \frac{i\pi}{2}$$

$$\frac{1}{2} 2 \operatorname{Ln}(2) - e^i - \frac{\pi}{2} i$$

$$\operatorname{Ln}(2) - \frac{\pi}{2} i - e^i$$

$$\operatorname{Ln}(2) - \frac{\pi}{2} i = (\cos(1) + \operatorname{sen}(1)i)$$

$$\boxed{\operatorname{Ln}(2) - (\cos(1) + (\operatorname{sen}(1) - \frac{\pi}{2})i)}$$

5. Sea $a \in \mathbb{R}$, $a \neq 0$. Determine $(AB)^T + 5C^{-1}$ si se tiene que:

$$A = \begin{pmatrix} -1 & 0 \\ 2 & 3 \\ 1 & 1 \end{pmatrix}, B = \begin{pmatrix} 0 & 2 & -\frac{3}{a} \\ 1 & -3 & -\frac{1}{a} \end{pmatrix} \text{ y } C = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & -2a \\ 0 & 1 & a \end{pmatrix}$$

$$AB = \begin{pmatrix} 0+0 & -2+0 & \frac{3}{a}+0 \\ 0+3 & 7-a & \frac{-6}{a} - \frac{3}{a} \\ 0+1 & 2-3 & \frac{-3}{a} - \frac{1}{a} \end{pmatrix}$$

$$AB = \begin{pmatrix} 0 & -2 & \frac{3}{a} \\ 3 & -5 & \frac{-9}{a} \\ 1 & -1 & \frac{-4}{a} \end{pmatrix}$$

$$(AB)^T = \begin{pmatrix} 0 & 3 & 1 \end{pmatrix}$$

